

Predictive Modeling and Explanatory Analysis of Instructor Performance Evaluations

Problem Statement

Student course evaluations are a critical metric for academic quality assurance and faculty development. Within the context of higher education, understanding the drivers of these evaluations is critical for instructional growth. The objective of this analysis is to model instructor rating outcomes, specifically the **Professor_Score** (the aggregated professor rating provided by their students), using a multi-dimensional data set including course characteristics, instructor attributes, enrollment composition, scheduling factors, and granular student evaluation dimensions.

This study adopts a dual-modeling approach to support both explanatory insight and operational decision-making:

- Explanatory Insight: Use **ElasticNet** to identify stable, directionally meaningful relationships between predictor variables and instructor ratings to inform academic governance.
- Predictive Accuracy: Use **XGBoost** to maximize predictive precision by capturing non-linear relationships, enabling proactive coaching and instructional support interventions.

Data Selection

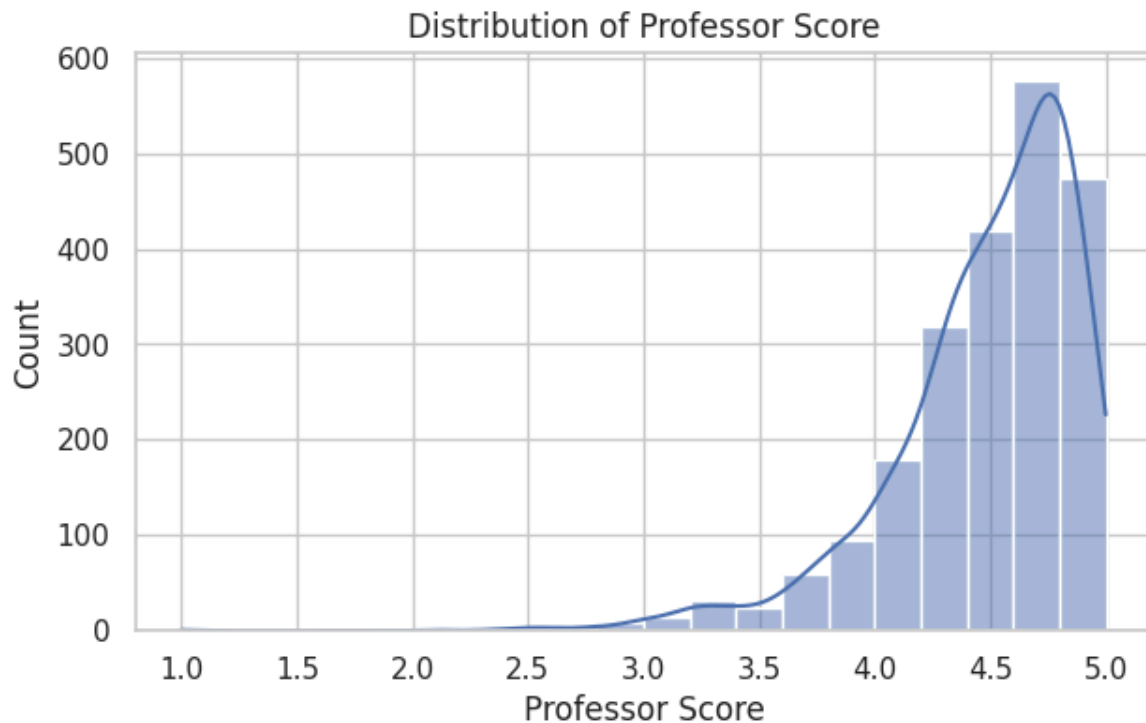
The dataset comprises course section-level data spanning two academic years and eight terms, totaling 2,208 records with 53 variables. The target variable, **Professor_Score**, is a continuous metric measured on a 1.0 to 5.0 scale. Key features were categorized and selected as follows:

- Instructor Attributes: Academic division, teaching status (FT/PT), and new faculty indicators.
- Course Structure: Credits, core vs. elective status, and introductory term markers.
- Operational Metrics: Enrollment totals (MS/PhD levels), waitlist counts, and instructional hours.
- Evaluation Dimensions: Standardized student perception questions regarding engagement, clarity, and inclusiveness.

Exploratory Data Analysis

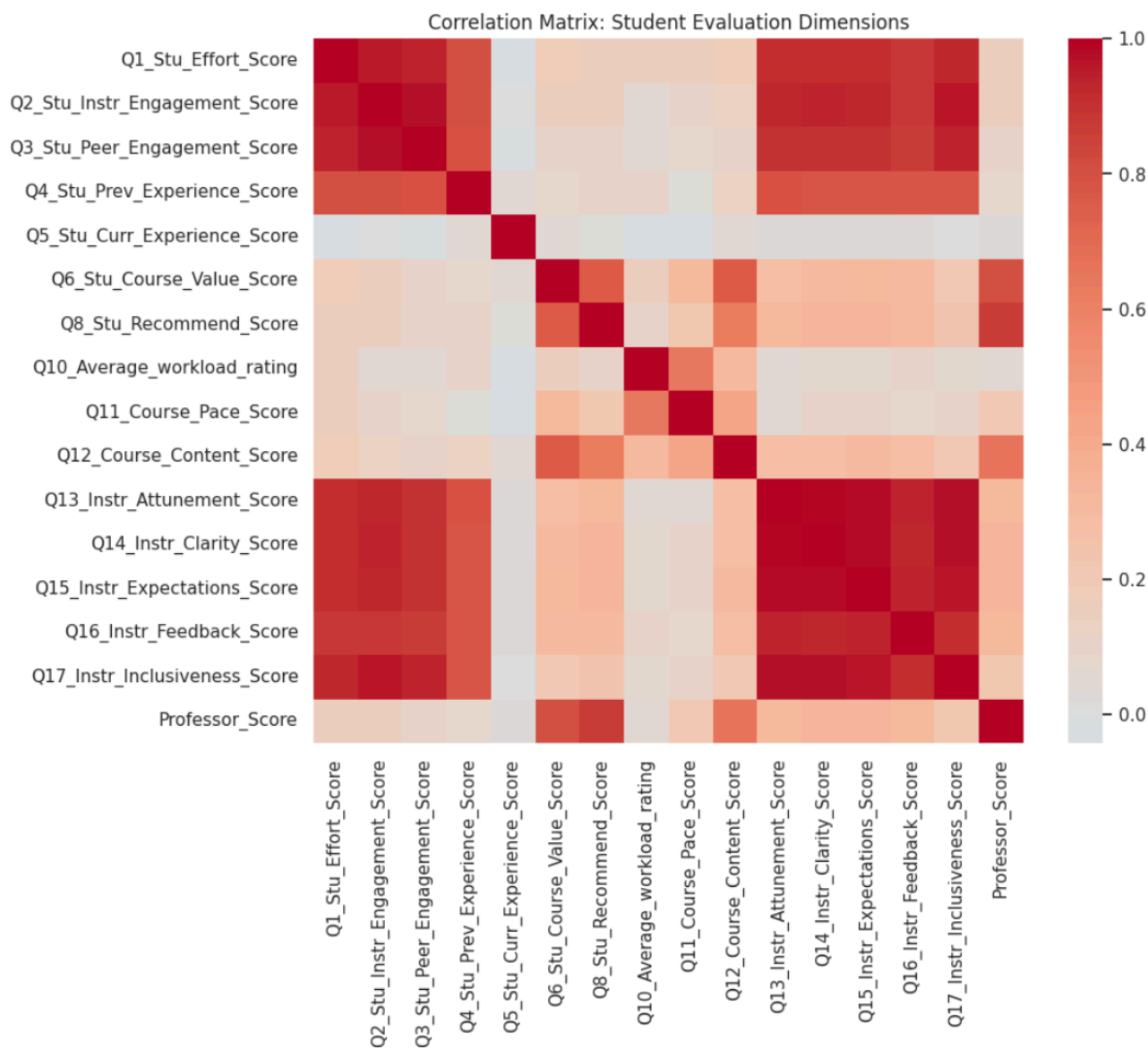
Mean Distribution of Target Variable Professor Score

Instructor rating scores are heavily concentrated at the upper end of the scale, with a mean exceeding 4.48 and a relatively narrow standard deviation of 0.41. This right-skewed distribution implies that most instructors are rated favorably; the modeling challenge is to distinguish between subtle differences in high performance rather than identifying extreme outliers.



Correlation Between Student Evaluation Dimensions

Correlation analysis of the student evaluation dimensions (Q1–Q17) revealed significant correlation with the final Professor_Score. The high correlation between student evaluation dimensions, such as instructional clarity and engagement, indicates significant multicollinearity. The high density of scores suggests that error metrics like Mean Absolute Error (MAE) provide better interpretability than R2 alone in this context.



Models Chosen

Two distinct modeling architectures were selected to fulfill the study's complementary goals:

- 1. **ElasticNet:** This model was chosen for its interpretability. It handles multicollinearity effectively while performing feature selection. This makes it ideal for identifying the primary, stable drivers of student satisfaction within a linear framework.
- 2. **XGBoost:** This gradient-boosted decision tree algorithm was chosen for its high-performance predictive capabilities. It is uniquely suited to capture complex interaction effects, such as how course workload might interact differently with core vs. elective courses that standard linear models often overlook.

Methods

The data was split into a term-based test set to simulate real-world academic cycles, ensuring the models were evaluated on their ability to predict outcomes for future terms based on historical data.

- For the ElasticNet model, the methodology prioritized feature stability and the mitigation of multicollinearity. To satisfy the model’s requirement for linear interpretability, high-cardinality categorical variables were removed to prevent the model from memorizing individual instances rather than learning generalizable trends. Furthermore, features underwent standardization to a common scale, and missing values in student evaluation dimensions were handled via median imputation to allow for regularization.
- The XGBoost methodology leveraged the algorithm's native ability to handle complex data structures. While still adhering to the term-based split and excluding pure identifiers to prevent data leakage, the XGBoost pipeline utilized one-hot encoding for categorical attributes and retained the inherent skewness of enrollment and capacity variables. The model was configured with 800 estimators and a learning rate of 0.05.

Results and Detailed Findings

Model Performance Comparison

Model	MAE	RMSE	R ²
Elastic Net (Interpretable)	0.0924	0.1230	0.8647
XGBoost (Predictive)	0.0722	0.1051	0.9012

ElasticNet Interpretation: Key Performance Drivers

The ElasticNet model identifies the specific variables that most reliably influence the Professor_Score. The model isolates factors with stable, directional impact, distinguishing between substantive instructional drivers.

Positive Drivers of Instructor Ratings

The analysis identified three primary contributors to high instructor evaluations:

- Student Recommendation (Q8): This emerged as the most dominant positive driver. The strong alignment between a student's willingness to recommend an instructor and the final score confirms that the evaluation is a reflection of overall student rating.
- Evaluation Clarity via Separate Evaluations: Interestingly, sections utilizing Separate Evaluations for instructors showed higher scores than those with joint evaluations. This suggests that students provide more favorable ratings when there is a clear, distinct attribution of instructional responsibility.
- Instructional Clarity (Q14) and Course Value (Q6) were identified as top-tier predictors. This indicates that ratings are driven less by rigid structural features and more by how effectively the instructor communicates complex content and establishes the relevance of the material.

Negative Drivers

While the magnitudes of negative factors were generally smaller than the positive drivers, they provide critical insights for administrative intervention:

- Co-Teaching Complexity: The presence of Co-Professors was the strongest negative indicator. This likely reflects the coordination challenges or perceived ambiguity in instructional delivery that can occur in team-taught environments.
- Higher Average Workload (Q10) and larger Course Capacity were associated with modest decreases in ratings. This suggests that as class size increases or the perceived workload intensifies, instructors face greater difficulty in maintaining the high level of individual engagement students expect.
- Peer Engagement (Q3) and Inclusiveness (Q17): These dimensions showed modest negative coefficients in the linear model. We assume this is primarily an artifact of multicollinearity; once clarity (Q14) and recommendation (Q8) are accounted for, these variables do not provide additional unique predictive power in a linear framework, though they remain critical components of the overall student experience.

XGBoost

Most Influential Predictive Factors

1. Program Degree: The specific academic program (e.g., Program_Degree_Masters1, Program_Degree_MSPHD) emerged as the most influential predictor. This suggests that student evaluative expectations for instructor performance, course rigor, and cohort dynamics differ

significantly across degree levels. This implies that a universal benchmark for faculty performance may be inappropriate without accounting for the specific program context.

2. Instructional Clarity and Expectation Setting: Consistent with the ElasticNet findings, Instructional Clarity (Q14) and Expectation Setting (Q15) remain dominant features. However, XGBoost identifies these as having nonlinear effects, suggesting that while incremental improvements in clarity help, there may be critical thresholds where poor communication of expectations causes a disproportionate drop in instructor ratings.
3. Student Recommendation and Perceived Course Value: The Student Recommendation Score (Q8) and Course Value (Q6) continue to be highly predictive.

Conclusion

The dual-modeling approach provides two distinct layers of utility for college leadership:

1. Strategic Governance (ElasticNet): The ElasticNet results indicate that 86.5% of the variance in ratings can be explained through a linear combination of attributes. This provides deans and department chairs with stable, interpretable evidence to support long-term faculty development. For instance, "clarity" as a key driver allows the university to prioritize specific communication workshops for all faculty.
2. Operational Proactivity (XGBoost): The XGBoost model's high accuracy ($R^2 = 0.901$) allows for real-time intervention. By predicting instructor scores before the final evaluation period ends, administrators can move from a reactive posture to a proactive one. If a section is predicted to fall below a quality benchmark based on preliminary metrics, administrators can implement targeted coaching or instructional support, directly improving the student experience before the term concludes.

While ElasticNet provides the evidence-based roadmap for long-term faculty development policy, XGBoost provides the tactical accuracy required for real-time instructional assessment. Together, these models empower the college to move from a reactive posture (reviewing end-of-term scores) to a proactive strategy that actively manages and improves the educational experience.